Publication-Level Holdout Separates Fingerprint and Excited-State Descriptor Models of Measured TADF Gaps

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Abstract

Do semi-empirical natural-transition-orbital (NTO) descriptors keep rank information on measured singlet–triplet gaps $\Delta E_{ST}$of thermally activated delayed fluorescence (TADF) emitters that structural fingerprints lose when molecules come from publications absent from training? In this corpus they do. On 231 donor–acceptor emitters (173 publications) from the Huang–Cole TADF database, random forests on GFN2-xTB/sTDA NTO descriptors order emitters across held-out publications with median Spearman $\rho = 0.34$, against 0.17 for Morgan fingerprints. Within a held-out paper, neither orders molecules. Neither has lower error: scaffold-cross-validated mean absolute errors are 0.091 eV (Morgan) and 0.096 eV (NTO), paired difference 95 % CI −0.017 to 0.009 eV, both 10 to 15 % below a mean-value predictor. NTO minus Morgan $\Delta\rho$ is 0.16 under source-paper holdout (central 95 % of 0.07 to 0.25 over 50 fold assignments); NTO ranks higher in 19 of 20 paper partitions, against 11 and 13 of 20 for matched scaffold and molecule partitions. NTO ρ barely moves across units, whereas Morgan ρ falls from 0.31 to 0.17 and RDKit 2D descriptors from 0.30 to 0.21. The contribution is this matched three-unit holdout comparison: fingerprint and RDKit ranking depends on same-publication molecules in training, NTO ranking does not. This is an association: siblings share structure and measurement protocol. Inter-report scatter caps R^2 near 0.80 to 0.86, above the observed 0.26. No laboratory transfer was tested. Here, random or scaffold splits overstated fingerprint ranking skill; source-publication holdout should accompany them.

Keywords: Thermally activated delayed fluorescence, singlet–triplet gap, molecular modelling, excited-state descriptors, machine learning, cross-validation

1. Introduction

1.1. Predicting the singlet–triplet gap of TADF emitters

Thermally activated delayed fluorescence (TADF) emitters—organic donor–acceptor molecules that harvest both singlet and triplet excitons for near-unity internal quantum efficiency [1]—are the heavy-metal-free route to tunable, efficient light sources for displays and for human-centric and horticultural lighting [2, 3]. An efficient emitter has to meet several requirements at once: a small singlet–triplet gap ($\Delta E_{ST} < 0.2$ eV), fast reverse intersystem crossing ($k_{RISC} > 1 \times 10^6\ s^{-1}$), and emission matched to the target band. This paper addresses only the first. The rate and device-efficiency requirements need spin–orbit-coupling and device-level calculations beyond the cheap structural data used here, and we make no quantitative claim about them (section 3.6).

1.2. Machine learning on computed and measured gaps

Designing such emitters means searching a large space of donor–acceptor combinations for a small ΔE_{ST}. The direct route—time-dependent DFT on excited states—scales steeply with system size [4], so exhaustive screening of more than a few hundred candidates is impractical. Machine learning offers a cheaper surrogate [5, 6], and several groups have predicted

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TADF gaps from molecular descriptors [7, 8]. Those low reported errors are obtained against *computed* (TD-DFT or higher) reference gaps. Nigam et al. [9] also worked with computed labels, gaps simulated at the ωB2PLYP′ level, and replaced an early regression model with a good/bad classifier (section 3.5); their study does not show how models behave on measured gaps. The closest prior work on measured gaps is that of He et al. [10], who trained machine-learning models on 1 321 experimental ΔE_{ST} records with measurement-condition features. Neither study compares a structural fingerprint with a semi-empirical excited-state representation under molecule, scaffold and source-publication holdout, which is the comparison made here. Experimental molecular-property learning is not, by itself, a new approach: JMGM studies have combined explainable predictors with experimental ligand–receptor observations [11] and modelled experimental infrared peaks [12]. Our question concerns the representation and the unit that is held out.

One problem recurs in this literature. Accuracy is often quoted against a single random train–test split, which rewards interpolation within familiar scaffolds and overstates how a model behaves on new chemistry. Literature-mined corpora sharpen this problem: one publication usually reports a series of close analogues measured under one protocol, so a random or scaffold split leaves same-paper siblings of a test molecule in training.

Our earlier work set the stage: Article 1 [13] benchmarked in JCIM GFN2-xTB/sTDA against 747 experimental molecules (MAE = 0.162 eV), xTB geometry relaxations, and scaffold cross-validation, and Article 2 [14] extracted structure–property design rules from that set. Here we ask: do semi-empirical natural-transition-orbital (NTO) descriptors retain rank information on measured TADF singlet–triplet gaps that structural fingerprints lose when molecules come from publications absent from training? The measured gaps come from the ChemDataExtractor-generated TADF database of Huang and Cole [15]; after matching to our descriptor set they cover 231 donor–acceptor emitters from 173 publications.

1.3. Our contributions

We compare error and rank ordering under three holdout units rather than claim a new high-accuracy predictor. The answer to the question is yes, within this corpus: under source-paper holdout the NTO model orders emitters across publications absent from training with median Spearman ρ = 0.34, against 0.17 for Morgan fingerprints. Neither representation orders molecules within a held-out paper, and neither has lower error than the other. The points below follow the order of the Results.

Fingerprint and NTO models have similar error.. On 231 molecules across 212 Bemis–Murcko scaffolds, random forests built from the 2D structure alone—Morgan fingerprints (MAE = 0.091 eV) and RDKit descriptors (0.100 eV)—match descriptors derived from a semi-empirical excited-state calculation (NTO spatial features, 0.096 eV); the Morgan and NTO models sit only 10 % to 15 % below a mean-value predictor (0.107 eV). The paired MAE difference is −0.0043 eV for Morgan minus NTO (95 % CI −0.0171 eV to +0.0092 eV). The error comparison does not move with the holdout unit: under source-paper holdout the median Morgan-minus-NTO difference is −0.0004 eV. Unlike a physics-guided model trained on NTO descriptors against a *computed* reference gap [16], we test the same class of descriptor against the *measured* gap.

Fingerprint ranking depends on same-publication molecules.. Over 50 random fold assignments, the NTO-minus-Morgan rank difference is $\Delta\rho$ = 0.160 under source-paper holdout (range 0.07 to 0.25), against 0.073 under scaffold and 0.055 under molecule holdout; NTO ranks higher in 19 of 20 repeated paper partitions, against 11 and 13 of 20 for matched scaffold and molecule partitions. The ranges describe fold-assignment variability, not confidence intervals. NTO ρ stays between 0.337 and 0.352 across units; Morgan ρ falls from 0.305 to 0.172. The drop sits on molecules with same-paper siblings: for them Morgan ρ is 0.54 when siblings are in training and 0.26 when the paper is held out, while on the 128 molecules from single-molecule papers it is 0.10 under either holdout. NTO is higher on those molecules, but the seed-averaged NTO-minus-Morgan contrast there ($\Delta\rho$ = 0.16, 95 % CI −0.05 to 0.37) is not resolved. RDKit 2D descriptors lose rank under paper holdout as Morgan does (ρ 0.303 to 0.209), so among the three sets tested the invariance belongs to the NTO descriptors. NTO orders emitters across publications absent from training; no representation orders molecules within a held-out paper. Siblings share close structure and measurement protocol, and these tests cannot separate the two, so we report the dependence as an association.

Label scatter leaves room the models do not use.. Independent experimental reports for the same compound scatter by 0.072 eV RMS (the largest within-molecule

standard deviation is 0.56 eV). For the 102 molecules with several reports, the median label has an RMS standard error of 0.049 eV; counting the 129 single-report labels at single-report scatter, label noise over all 231 molecules is about 0.072 eV RMS. Individual discrepancies in reported ΔE_{ST} have been noted for TADF emitters [17]; here we quantify inter-report variability across the 231-molecule corpus drawn from 173 papers. Applying the noise-ceiling framework of Crusius et al. [18] with per-molecule label noise gives $R^2_{max} \approx 0.80$–0.86 (section 3.4), well above the observed $R^2 \approx 0.26$, so label noise is not the dominant limit and it does not establish a 0.1 eV MAE floor. Computed references face a separate limit: B3LYP and CAM-B3LYP gaps differ by up to 0.58 eV. On this corpus neither a higher-capacity tabular learner (gradient boosting, MLP, SVR) nor the one frozen pretrained molecular-transformer embedding we tested beats the random forest (the latter reaches MAE 0.109 eV to 0.113 eV). Low-gap enrichment on held-out labelled data (1.2 to 1.4× over random) and a filtered virtual library are reported in the SI; neither supports molecule-level selection, and we make no device-level performance claims.

This study builds on our earlier work—**Article 1** [13] (xTB/sTDA validation) and **Article 2** [14] (design rules)—by testing how the holdout unit changes the comparison on measured labels. The measured labels come from the Huang–Cole TADF database [15]. Data, code and the analysis ledger for the three related articles are archived under Zenodo concept DOI 10.5281/zenodo.17436069 (latest version) and mirrored on GitHub; the scripts and outputs of the holdout analyses reported here are in that archive and are also supplied as Supplementary Data (see the Data and code availability statement).

The contribution of this work is a matched comparison of three holdout units (molecule, scaffold and source publication) on the same 231 measured gaps, which shows that fingerprint and RDKit ranking of measured TADF gaps depends on same-publication molecules in training while semi-empirical NTO ranking does not. The error null carries its own information. Because the NTO and fingerprint models do not differ in error, the sTDA calculation is not justified by accuracy; its measurable value is ranking across publications. The 10 % to 15 % gain over the mean-value predictor is the margin a new structure representation has to beat on these labels.

2. Methods

2.1. Experimental target and dataset

The prediction target is the *experimental* singlet–triplet gap ΔE_{ST}. We took literature-reported values from the ChemDataExtractor-generated TADF database of Huang and Cole [15] and matched them, by compound name, to molecules in our descriptor set, retaining values in the physical range −0.3 eV to 1.5 eV. Where a compound carried multiple independent reports we used the median and recorded the inter-report spread; a single report sits 0.072 eV from the median (RMS, weighting each molecule equally; the report-pooled value is 0.0905 eV). For the 102 multi-report molecules the median carries an RMS standard error of 0.049 eV; with the 129 single-report labels counted at single-report scatter, the label noise over all 231 molecules is about 0.072 eV RMS (section 3.4). The final benchmark comprises 231 donor–acceptor molecules spanning 212 distinct Bemis–Murcko scaffolds [19].

Each molecule was assigned one *primary DOI*: the source publication that contributes the most reports for that molecule (ties are broken by the code's report-count ordering). This gives 173 primary-DOI groups for the 231 molecules, of which 128 contain one molecule and 45 contain more than one (`code/jmgm_doi_group_ranking_check.py`).

The corpus does not record measurement medium (solvent, host) for individual values, so the per-molecule median is a consensus over uncontrolled conditions rather than a single, solvent-specific state. Because environmental electrostatics and conformational disorder are recognised as primary obstacles to accurate TADF modelling,[20] this consensus is best read as the central tendency of experimentally accessible S_1–T_1 gaps; it trades physical specificity for reduced variance. For the 102 molecules with repeat reports, a single random report deviates from this median by 0.072 eV on average—comparable to the model error itself (Table S3). For those molecules the median is known more precisely, to 0.049 eV RMS standard error.

The sTDA S_1 and T_1 energies are excluded from the NTO set because supplying them reproduces a computed gap exactly: under scaffold cross-validation a linear model given both energies recovers $E_{S_1} - E_{T_1}$ with $R^2 = 1.000$, and a random forest reaches $R^2 = 0.59$ on the same features (Section S3.1b).

2.2. Cheap electronic-structure inputs and physics-informed descriptors

Excited-state quantities used to build descriptors were obtained at the semi-empirical level: GFN2-xTB [21] geometries (`xtb` version 6.7.1) followed by sTDA [22] for the lowest singlet and triplet states. From the natural transition orbitals (NTOs) [23] of each excitation we computed spatial overlap descriptors that encode exchange-integral physics:

$$S_{\mathrm{he}} = \int \phi_{\mathrm{hole}}(\mathbf{r})\, \phi_{\mathrm{electron}}(\mathbf{r})\, d\mathbf{r}, \tag{1}$$

$$\Delta\lambda_{\mathrm{A}} = |\lambda_{\mathrm{A}}(S_1) - \lambda_{\mathrm{A}}(T_1)|, \tag{2}$$

$$\mathrm{Char}^2_{\mathrm{diff}} = \Big[\textstyle\sum_i |q_i(S_1) - q_i(T_1)| \Big]^2, \tag{3}$$

$$\Delta S_{\mathrm{he}} = |S_{\mathrm{he}}(S_1) - S_{\mathrm{he}}(T_1)|. \tag{4}$$

S_{he} quantifies hole–electron overlap (related to the exchange integral), $\Delta\lambda_{\mathrm{A}}$ acceptor-participation differences, $\mathrm{Char}^2_{\mathrm{diff}}$ charge-transfer character changes, and ΔS_{he} the singlet–triplet overlap difference. The full set of 35 NTO descriptors (the S_1/T_1 energy scalars excluded) is listed in the Supporting Information; these descriptors require the sTDA excited-state calculation, even though they omit the energies themselves. The ground-state wavefunction for sTDA was generated with `xtb4stda`, and excitations were computed with `stda -xtb -e 10` (sTDA-xTB parametrisation, 10 eV excitation-energy window; `-t` for triplets), all in the gas phase; descriptors use the sTDA rather than the sTD-DFT (`-rpa`) results.

2.3. Feature sets and machine-learning models

We compared three feature representations on the same molecules. Two require no quantum chemistry, being computed from the 2D structure alone: (i) RDKit physicochemical descriptors [24] and (ii) Morgan fingerprints (radius 2, 2 048 bits; RDKit `GetMorganFingerprintAsBitVect` with its defaults, i.e. no chirality and no feature invariants). The third, (iii) the NTO spatial descriptors above, is derived from the sTDA excited state but excludes the energy scalars. A reference set that additionally includes the sTDA S_1/T_1 energies was also evaluated. We used random-forest regressors [25] (400 trees, scikit-learn [26]), which outperformed support-vector regression, ridge regression and a mean-value baseline on this task.

Evaluation.. Error metrics, table 1, the baselines of Table S22, the SHAP analysis and the label-noise analyses use Bemis–Murcko scaffold `GroupKFold` ($k = 5$); the source-paper (DOI-grouped) results and the holdout-unit controls below use the holdout unit stated with each number. Because 92.9 % of scaffolds are singletons, this split is numerically close to a random split; we therefore also report five-fold source-paper (DOI-grouped) out-of-fold predictions. That grouping holds out source publications and any synthetic series or measurement practices correlated with them, but does not identify a laboratory or control the protocol (section 3.1). The corpus records no solvent, host or temperature per value, so a paper DOI is only a proxy for shared conditions, not a measured protocol. We benchmark against a mean-value predictor, the raw sTDA gap, and a range-separated TD-DFT reference gap (section 3.1), and report 95 % confidence intervals from 2 000 bootstrap resamples of the out-of-fold predictions. For the DOI-grouped rank contrast, we also resampled the 173 primary DOI groups as clusters (5 000 paired percentile-bootstrap draws, seed 54 321) while keeping the original out-of-fold predictions fixed. As a partition sensitivity check, we refitted both random forests in 20 source-paper `GroupShuffleSplit` partitions (20 % of DOI groups held out; seed 17) and computed Spearman correlation on each test partition. These overlapping partitions are descriptive, not independent confirmations or a confidence interval for new laboratories.

Holdout-unit controls.. To separate the effect of the holdout unit from the effect of the error-versus-ranking metric, we compared three units on the same 231 molecules: single molecule (random folds), Bemis–Murcko scaffold (212 groups) and primary source publication (173 groups). For each unit we drew 50 random assignments of groups to 5 folds, fitted both random forests (400 trees) on each assignment and computed pooled out-of-fold Spearman ρ and MAE for Morgan and NTO features. The spread over the 50 assignments (2.5th–97.5th percentiles) describes fold-assignment variability; it is not a confidence interval. The 20-partition `GroupShuffleSplit` count was repeated for each unit. On a fixed scaffold split we also resampled scaffold clusters (5 000 draws). The same 50 assignments were reused to split the molecules into 103 from papers with at least one other molecule in the corpus (multi-molecule papers) and 128 from single-molecule papers, and to compute (i) ρ within each subset and (ii) a sibling-pull statistic, the Spearman correlation between a molecule's out-of-fold prediction and the leave-self-out mean label of the other molecules from the same paper (103 molecules). Code: `code/a017_split_unit_controls.py` and `code/a019_series_shortcut_test.py`; outputs

in `data/a017_split_unit_controls.json` and `data/a019_series_shortcut_test.json`. The same 50 molecule- and paper-holdout assignments were repeated with the 217 RDKit 2D descriptors, and within-paper ranking was measured as the Spearman correlation between paper-centred labels and paper-centred predictions on the 103 multi-paper molecules. For the 128 single-paper molecules, out-of-fold predictions were averaged over the 50 paper-holdout assignments and the NTO-minus-Morgan ρ difference was given a paired molecule bootstrap interval (5 000 draws, seed 54 321); code `code/a020_controls.py`, output `data/a020_controls.json`. These analyses (`a017`–`a020`) were designed after the fixed-split result (`a016`) had been seen; the decision rule of each was fixed before it was run and is recorded in the script's docstring. Code and full per-split values for the fixed split are in `code/jmgm_doi_group_ranking_check.py` and `data/a016_jmgm_doi_group_ranking.json`.

TD-DFT reference gap.. For a higher-level reference we computed vertical ΔE_{ST} for all 231 molecules at wB97X-D4/def2-SVP with RIJCOSX and the Tamm–Dancoff approximation (ORCA 6 [27]; `nroots` 8, triplets included; GFN2-xTB ground-state geometry). All 231/231 jobs converged (76.2 h elapsed on 8 processes, so 609 core-hours; median 16.1 min per molecule, slowest 2.1 h). These are measured wall times from `results/tddft_reference_231.csv`, not a scaling extrapolation. The raw gap, an in-fold linearly calibrated gap, and the gap as an added descriptor in the random forest model are evaluated against the mean-value baseline on the same scaffold and DOI cross-validation folds (section 3.1; Table S17). B3LYP/6-31G(d) and CAM-B3LYP/def2-TZVP calculations on 7 emitters (functional dependence of the computed gap), and CAM-B3LYP/def2-TZVP calculations on 14 emitters (adiabatic gaps and the T_1–T_2 manifold) are described in Section S4; the substructure-filtered PubChem library is described in Section S5.

Interpretability and reproducibility.. Feature importance was quantified with exact TreeSHAP [28] on the fitted forest. All scripts (`code/finalize_model.py`, `code/rebuild_structural_model.py`) regenerate every reported number and fig. 1. Software: scikit-learn 1.8.0 and RDKit 2026.03.2. Every random forest used 400 trees, `random_state = 0` and otherwise scikit-learn defaults. The 50 fold assignments per holdout unit were drawn with NumPy generator seeds 0, 1, …, 49 (one seed per assignment, `numpy.random.default_rng(seed)`); the paper-cluster, scaffold-cluster and molecule bootstraps used seed 54 321, and the 20 `GroupShuffleSplit` partitions used seed 17. Data source, splits, features, model, hyperparameters, metrics and code location are collected in Table S24.

Bootstrap confidence intervals and power analysis.. Point estimates of the rank correlation are complemented by 5 000-replicate percentile bootstrap confidence intervals on the out-of-fold predictions (`code/r2_ci_power_analysis.py`). For the headline model the Spearman $\rho = 0.36$ ($n = 231$) has a 95 % CI of 0.24 to 0.47 ($p = 2.6 \times 10^{-8}$) and the Pearson $r = 0.50$ a 95 % CI of 0.26 to 0.68; both exclude zero, establishing statistically significant association despite the wide R^2 interval. A power analysis for the correlation test (Fisher z transform, two-sided, $\alpha = 0.05$) shows that detecting $\rho = 0.36$ at 80 % power requires only $n \approx 60$ (90 % power: $n \approx 79$); at $n = 231$ the achieved power is ≈ 1.0. This bears on the ranking question only: it establishes that the ρ estimate is not sample-size-limited, and it says nothing about the width of the R^2 interval, which is a different statistic and is discussed on its own terms in section 3.1. We report Spearman ρ as the primary metric for triage utility because triage is a ranking task, not because it is the more favourable number; the full set of metrics and intervals is collected in Table S6.

2.4. Data and code availability

Labels: Huang–Cole TADF database [15]. Data, code and the full analysis ledger for the three related articles: Zenodo concept DOI 10.5281/zenodo.17436069 (latest version), mirrored on GitHub (Data_Articles_TADF). The holdout-analysis scripts and outputs (A-016 to A-020) are in that archive (extttML_reproducibility/code, extttML_reproducibility/data) and are also enclosed as Supplementary Data (extttsupplementary_data/, extttJMGM_group_ranking_supplement.zip). File list: Data and code availability statement after the Conclusions.

3. Results and Discussion

3.1. Under scaffold holdout, structure-based models sit 10 % to 15 % below the mean-value error

We assembled 231 donor–acceptor molecules for which an experimental ΔE_{ST} could be matched to a structure, and evaluated random-forest regressors under Bemis–Murcko scaffold cross-validation (212 distinct

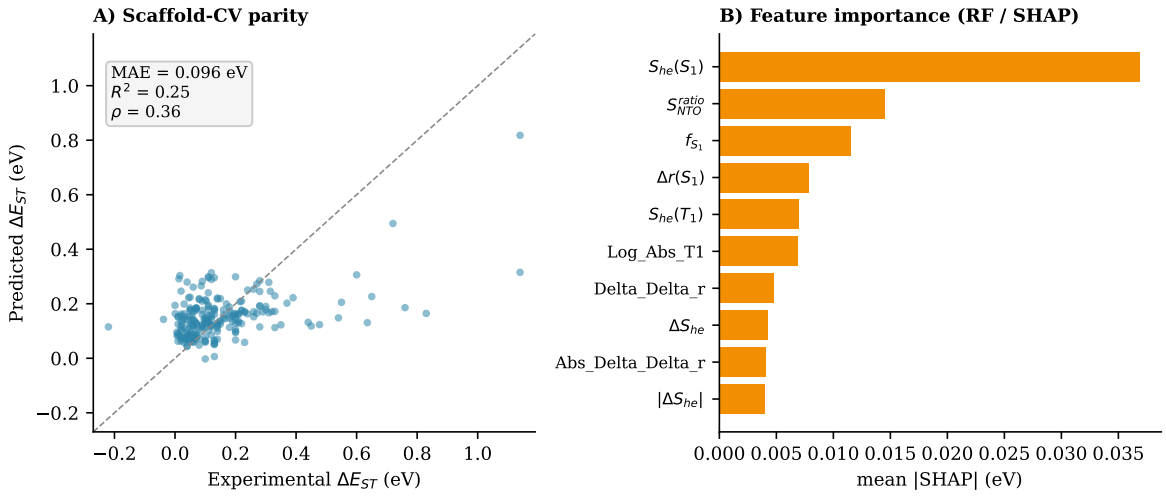


Figure 1: Structure-based prediction of experimental ΔE_{ST}. (A) Scaffold cross-validated parity for the random forest on NTO spatial descriptors (231 molecules, 212 Bemis–Murcko scaffolds): MAE = 0.096 eV, R^2 = 0.25, Spearman ρ = 0.36. Models built from 2D structure alone (Morgan, RDKit) have similar errors (table 1); formal equivalence was not established. The model regresses high-gap outliers toward the mean, the expected behaviour when the training distribution is dominated by low-gap emitters. (B) SHAP importances: hole–electron overlap S_{he}, NTO overlap ratio, oscillator strength f_{S_1} and charge-transfer distance Δr dominate, consistent with exchange-integral physics.

scaffolds), so that no scaffold appears in both training and test. Because 92.9 % of scaffolds are singletons, scaffold and random splits produce numerically similar folds (test–train nearest-neighbour Tanimoto differs by only 0.022). The scaffold split guarantees zero shared scaffolds per fold (versus 3 to 8 under a random split) but task difficulty barely changes; the harder test of transfer is the source-paper split reported below. That the model learns real signal rather than scaffold-group structure is confirmed by a label-permutation test: the true scaffold-CV MAE (0.096 eV) lies far below the null from 1 000 target shuffles (mean 0.121 eV, p = 0.001; Table S11). The headline accuracy is also stable across validation schemes: replacing the singleton scaffolds with genuine multi-molecule Butina clusters (down to 70 groups, largest 38 molecules) leaves the MAE at 0.099 eV to 0.104 eV, between scaffold-CV (0.096 eV) and random-CV (0.106 eV) while preserving the rank correlation (Spearman ρ spanning 0.31 to 0.40 across cutoffs), showing that the ranking capability does not degrade under aggressive clustering (Table S19). The three feature sets perform within 0.009 eV of one another (table 1); for the fingerprint and excited-state models, which we compare head to head below, that closeness is backed by a paired bound of 0.017 eV. Two are computed from the 2D structure alone, with no quantum chemistry: Morgan fingerprints (MAE = 0.091 eV) and RDKit physicochemical descriptors (0.100 eV). The third, NTO spatial descriptors, is derived from a semi-empirical sTDA excited-state calculation but excludes the S_1/T_1 energy scalars; it reaches MAE = 0.096 eV (95 % CI 0.082 to 0.109), R^2 = 0.25 (CI 0.03 to 0.45), Spearman ρ = 0.36. That the tested structure-only sets have similar errors to the NTO set is the central comparison: the semi-empirical calculation has no measured error advantage on these folds. Adding back the S_1/T_1 energies does not improve the forest's MAE either (table 1). The *Energies + NTO* set has MAE 0.097 eV versus 0.096 eV for NTO alone. Similar predictive errors do not establish that the feature sets encode identical physical information. The natural comparator is the mean-value baseline (MAE = 0.107 eV), over which every model improves by only about 10 %; the large apparent gain over the raw sTDA gap (0.25 eV) reflects that the semi-empirical gap is badly biased, not that the model is accurate. The R^2 confidence intervals are wide and, for the structure-only sets, include negative values (RDKit −0.11 to 0.51; Morgan −0.19 to 0.53): at 95 % confidence we cannot exclude that these models are no better than predicting the mean. The signal is real but weak.

A higher level of theory does not change this picture. A range-separated TD-DFT gap (wB97X-D4/def2-SVP, Tamm–Dancoff, 609 core-hours) overestimates the measured gap by a systematic 0.65 eV and, used directly, is the least accurate predictor in table 1 (MAE 0.654 eV, R^2 = −16.3). Its correlation with experiment is higher than the semi-empirical gap's (Pear-

son r from 0.32 to 0.43), and an in-fold linear calibration that removes the offset brings it to MAE 0.099 eV—level with the mean-value baseline, still short of the structure-only forest. Added as a single feature it lowers the Morgan model from 0.091 to 0.087 eV (paired 95 % CI 0.0004 to 0.0071, Wilcoxon $p = 0.016$): a small gain for 609 core-hours (Table S17). This higher-level TD-DFT descriptor is distinct from the semi-empirical NTO spatial set: the latter does not lower the scaffold-CV error, whereas the former yields a measurable but modest improvement.

Holding out whole source papers rather than scaffolds tests a different axis: it removes entire publications and correlated chemical series. Because the corpus records no measurement conditions per value, this is a proxy for shared practices, not a laboratory- or protocol-controlled experiment. Under DOI-grouped five-fold `GroupKFold` (173 papers, 231 molecules) the Morgan model reaches MAE 0.102 eV ($R^2 = 0.18$; NTO $R^2 = 0.142$), and its paired advantage over the mean-value baseline falls to 0.005 eV with a 95 % CI that crosses zero. On accuracy, the model's advantage is not resolved under this source-paper split.

Ranking behaves differently in these same folds: on this fixed fold assignment the NTO model has Spearman $\rho = 0.35$ while Morgan has 0.16 (MAE 0.099 eV against 0.102 eV). Their paired $\Delta\rho = 0.19$ (the fixed-split value; the median over 50 random fold assignments is 0.16, section 3.3) has a molecule-resampled 95 % CI of 0.04 to 0.34. Resampling the 173 primary source DOIs instead gives 0.014 to 0.366; that paper-cluster interval barely excludes zero and remains conditional on these fitted out-of-fold predictions. In 20 additional paper-group partitions, retraining both models with 20 % of DOI groups held out each time, NTO has the higher test-set rank correlation in 19 (median $\Delta\rho = 0.134$; range −0.014 to 0.404). Those overlapping same-corpus partitions show sensitivity to grouping rather than independent replication. Together these tests identify a possible NTO ranking use case, not an advantage established for selection in a new laboratory. One observation bounds the pessimistic reading, with a caveat that must travel with it: 58 % of repeatedly-measured molecules have zero inter-report spread, and on the 102-molecule multi-report subset the model reaches the study's lowest MAE, 0.080 eV. That subset is also an easier target: a mean-value predictor restricted to it scores 0.089 eV, so the model's advantage there is 0.009 eV with a 95 % CI spanning zero. The low absolute error reflects the narrower spread of those labels more than any gain in prediction.

For the NTO variant, SHAP (SHapley Additive exPlanations) analysis (fig. 1B) identifies hole–electron overlap S_{he} (which governs the magnitude of the exchange interaction), the NTO overlap ratio (tracking transition purity), oscillator strength f_{S_1} (a proxy for transition dipole coupling), and the charge-transfer distance Δr (which dictates spatial separation) as the dominant predictors—physical quantities that jointly constrain the exchange integral governing ΔE_{ST}. This ranking is stable rather than an artefact of one fit: across 200 bootstrap refits the leading descriptor $S_{\mathrm{he}}(S_1)$ stays in the top five 99.5 % of the time, and on average 3.7 of the five canonical descriptors recur (Table S10). Restricting the attribution to the reliable-prediction subset—the 152 molecules whose out-of-fold residual is below 0.10 eV—leaves $S_{\mathrm{he}}(S_1)$ ranked first in 99.5 % of refits, so the leading attribution does not rest on the poorly-predicted tail. An independent, model-agnostic permutation-importance analysis ranks the same descriptor, $S_{\mathrm{he}}(S_1)$, first, so the leading attribution is corroborated by two methods (the lower-ranked tail is not; SI section S6.3). Given the model's modest ranking power, we read only the dominant attribution as a mechanistic hypothesis for experimental test rather than a validated causal claim; the structure-only fingerprint and descriptor models reach the same accuracy without any excited-state calculation.

3.2. NTO and fingerprint models have similar error under scaffold holdout

Do quantum-mechanical features beat pure structural descriptors when the target is an experimental property? In scaffold cross-validation the tested NTO representation does not reduce error. Paired on identical folds, the NTO model (MAE = 0.096 eV) and the Morgan model (0.091 eV) differ by ΔMAE = −0.0043 eV for Morgan minus NTO (95 % CI −0.0171 to +0.0092, Wilcoxon $p = 0.66$). The lower confidence limit is close to the proposed 0.017 eV equivalence margin; we report the observed similarity rather than claim a formal equivalence test. Both outperform the mean-value baseline in paired comparisons, but neither predicts individual gaps with the precision needed for emitter selection.

The accuracy comparison does not settle ranking on source-paper holdouts. In the DOI-grouped folds (section 3.1), NTO retains Spearman $\rho = 0.35$ while Morgan has $\rho = 0.16$; their paired difference on this fixed fold assignment is $\Delta\rho = 0.19$ (paper-cluster 95 % interval 0.014 to 0.366), and the median over 50 fold assignments is 0.16 (section 3.3). In a separate scaffold-CV triage analysis, at the top 20 % of the labelled benchmark NTO enrichment is 1.6-fold versus 1.4-fold for

Table 1: Prediction of experimental ΔE_{ST} under scaffold cross-validation (random forest, 231 molecules); R^2 with 95 % bootstrap CI. The 2D-structure-only sets (no quantum chemistry) match the NTO set, which requires an sTDA excited-state run; none approaches the precision a quantitative photophysical method would require, and the structure-only R^2 intervals include zero. The range-separated TD-DFT reference gap (wB97X-D4/def2-SVP, Tamm–Dancoff; 609 core-hours) is listed raw, after an in-fold linear calibration, and as an added feature (Table S17).

Feature set	MAE (eV)	R^2 (95 % CI)	Spearman ρ
Morgan FP (2D structure only)	0.091	0.27 (−0.19, 0.53)	0.31
RDKit desc. (2D structure only)	0.100	0.31 (−0.11, 0.51)	0.26
NTO spatial (needs sTDA)	0.096	0.25 (0.03, 0.45)	0.36
Energies + NTO (reference)	0.097	0.24	0.34
Mean-value baseline	0.107	0.00	—
Raw sTDA gap	0.251	−2.22	0.24
Physics reference gaps (higher level of theory)			
Raw TD-DFT gap (wB97X-D4/def2-SVP)	0.654	−16.3	0.38
Calibrated TD-DFT gap (in-fold)	0.099	0.17	0.37
Morgan FP + TD-DFT gap	0.087	0.31	0.37

Morgan. The next subsection tests whether this ranking contrast comes from the evaluation criterion or from the holdout unit. It does not establish that the contrast persists on independent corpora or controlled protocols. Recent theoretical work by Salvi et al. [29] on electronic correlation suggests a possible limitation of semi-empirical descriptors; the present comparison alone cannot identify its cause. The two representations differ in what they can be read for, though not in a way the attribution totals establish on their own. The ten highest-attribution NTO descriptors carry 79 % of that model's SHAP weight against 30 % for the ten highest-attribution Morgan bits (Figure S2); those totals are not comparable, because ten descriptors are 29 % of the 35-dimensional NTO set and ten bits are 0.5 % of a 2 048-bit fingerprint, so the concentration follows from dimensionality. What does not follow from dimensionality is that the NTO features are *named* donor–acceptor overlap and charge-transfer quantities, whereas Morgan bits are anonymous hashed substructures. The case for the physics-informed representation is interpretability of the variables themselves, not a larger share of attribution.

3.3. Fingerprint ranking depends on same-publication molecules

The holdout unit, not the error-versus-ranking criterion, sets the size of the NTO–Morgan rank contrast. We repeated the comparison with 50 random assignments of groups to five folds for each of three units (table 2). Under source-paper holdout the median $\Delta\rho$ (NTO minus Morgan) is 0.160 and is positive in 50 of 50 assignments; under scaffold holdout it is 0.073 (49/50 positive) and under random molecule holdout 0.055 (45/50 positive). The 0.191 of the fixed split lies in the upper part of the paper-holdout distribution, so the fold-assignment-robust value is ≈ 0.16 (50-assignment range 0.07 to 0.25); the ranges describe fold-assignment variability, not confidence intervals. The error comparison does not move with the unit: Morgan minus NTO MAE is −0.0004 eV under paper holdout and −0.0045 eV under scaffold holdout, with ranges that span zero, and −0.0081 eV under random holdout, where Morgan is slightly better (range −0.0157 to −0.0002). NTO is higher in 19 of 20 repeated paper-group partitions, against 11 of 20 for scaffold and 13 of 20 for molecule partitions, so the earlier 19/20 count is specific to the paper unit.

Table 2: Rank correlation of random-forest models by holdout unit (same 231 molecules; 50 random group-to-fold assignments, medians). $\Delta\rho$ is NTO minus Morgan; the range is the 2.5th to 97.5th percentile over assignments, a measure of fold-assignment variability, not a confidence interval. The last column counts 20 repeated holdout partitions in which NTO has the higher test-set ρ. RDKit 2D descriptors were run under molecule and paper holdout only (n.t., not tested); their multi-paper ρ falls from 0.518 to 0.350 and single-paper ρ under paper holdout is 0.102.

			median ρ			NTO
Unit (groups)	$\Delta\rho$	Range	Morgan	RDKit	NTO	higher
Paper (173)	0.160	0.073–0.246	0.172	0.209	0.337	19/20
Scaffold (212)	0.073	0.007–0.144	0.269	n.t.	0.351	11/20
Molecule (231)	0.055	−0.033–0.126	0.305	0.303	0.352	13/20

The contrast comes from the fingerprint side. NTO rank correlation is nearly invariant to the unit (median ρ 0.337 to 0.352), whereas Morgan ρ falls from 0.305

(random) to 0.269 (scaffold) to 0.172 (source paper) (fig. 2a). Splitting the molecules by paper multiplicity locates the drop (fig. 2b). For the 103 molecules whose paper contributes at least one other molecule, Morgan ρ is 0.54 when siblings are in training and 0.26 when the whole paper is held out, while NTO moves from 0.48 to 0.45. For the 128 molecules from single-molecule papers, Morgan ρ is 0.10 under either holdout against 0.24 for NTO; with predictions averaged over the 50 paper-holdout assignments the two are 0.11 and 0.27, a difference of 0.16 whose paired bootstrap 95 % CI (−0.05 to 0.37) includes zero, so this subset contrast is not resolved. The sibling-pull statistic, the Spearman correlation between a molecule's prediction and the mean label of its same-paper siblings, is 0.68 for Morgan under random holdout and 0.26 under paper holdout, against 0.47 and 0.42 for NTO. Morgan's MAE on multi-paper molecules is 0.025 eV lower under random than under paper holdout (range 0.014 to 0.034); NTO's difference is 0.007 eV, with a range that includes zero.

A dense descriptor set without quantum chemistry behaves like the fingerprint. The 217 RDKit 2D descriptors, run through the same 50 assignments, reach median ρ 0.303 under random and 0.209 under paper holdout; on multi-paper molecules ρ falls from 0.518 to 0.350, and on single-paper molecules it is 0.102 (table 2). Among the three sets tested, holdout-unit invariance is specific to the NTO descriptors, and dimensionality alone does not explain it. None of the three orders molecules *within* a held-out paper: the Spearman correlation between paper-centred labels and paper-centred predictions on the 103 multi-paper molecules is −0.06 to 0.03 under paper holdout. The NTO advantage is in ordering emitters across publications absent from training. Under random holdout Morgan's within-paper ρ is −0.35: its predictions are pulled towards the sibling mean, which inverts the order inside a paper.

This is an association. Molecules from one paper share close structure and measurement protocol, and these tests cannot separate structural near-duplication from laboratory or protocol effects. The paper-multiplicity test compared random with source-paper holdout only (scaffold holdout was not part of it): the fingerprint model's ranking skill under random holdout sits mostly on molecules whose same-paper siblings are in training, and the NTO descriptors keep their rank information when the siblings are absent. All units use one corpus and 128 of the 173 papers contribute a single molecule, so paper holdout differs from random holdout through the 45 multi-molecule papers only; no test here crosses laboratories.

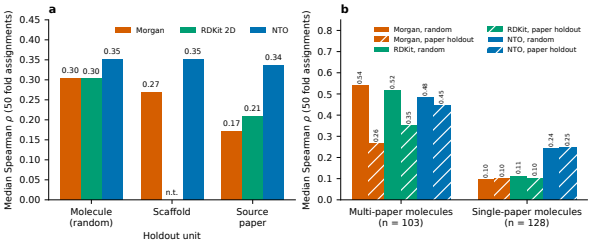


Figure 2: Rank correlation by holdout unit and by paper multiplicity (medians over 50 random fold assignments; random-forest models on the same 231 molecules). (a) Spearman ρ for Morgan, RDKit 2D and NTO features under molecule, scaffold and source-paper holdout (RDKit was not run under scaffold holdout; n.t.). (b) ρ for the 103 molecules from multi-molecule papers and the 128 from single-molecule papers, under random and paper holdout. The full seed-range tables are in the SI.

3.4. *Label scatter leaves the ceiling at $R^2_{max} \approx 0.8$, well above the observed R^2*

The observed MAE is near 0.1 eV, but label variability does not by itself account for the model's error. Independent experimental reports for the same compound disagree substantially: a single randomly chosen report sits 0.072 eV from its molecule's median (RMS with each molecule weighted equally), the disagreement exceeds the model MAE for 19.6 % of molecules, and the worst molecule reaches a standard deviation of 0.56 eV; the eight independent values for 4CzIPN alone span 0.01 eV to 0.48 eV (Table S3). This scatter is consistent with uncontrolled measurement conditions (solvent polarity, concentration, host matrix), which influence photophysical measurements [30]; the corpus cannot separate conditions, protocol and transcription error. That ΔE_{ST} is reported inconsistently has been shown for individual emitters [17]; quantified across 231 molecules and 173 source papers it provides an inter-report variability estimate for this TADF corpus. This variability constrains any predictor fitted to the measured labels. The quantity the model is regressed on, however, is the per-molecule *median*. For the 102 molecules with several reports, the RMS standard error of that median is 0.049 eV; the other 129 labels rest on one report each, and counting them at single-report scatter gives an RMS label noise of about 0.072 eV over all 231 molecules. Label imprecision is therefore a real term in the error budget but does not by itself account for it, and we do not claim the forest sits at a noise floor. Applying the noise-ceiling framework of Crusius et al. [18] ($R^2_{\max} = 1 - \sigma^2_{\text{noise}}/\sigma^2_y$) with per-molecule noise (the standard error of the mean for the 102 multi-report molecules, and the single-report scatter for the 129 single-report molecules) gives $R^2_{\max} \approx 0.81$

(95 % CI 0.68 to 0.88) for the pooled within-molecule SD of 0.0856 eV; the alternative single-report estimates give 0.80 and 0.86. All lie well above the observed $R^2 \approx 0.26$. The model captures about 30 % to 32 % of that ceiling, so label noise alone is not the dominant constraint. The assumptions behind the ceiling are listed in SI section S3.3b. Progress requires both a better dataset *and* improvements in features or corpus scale; label noise (about 0.072 eV RMS over the 231 molecules) is the term a curated benchmark would directly attack. A second, separate dispersion applies to computed references rather than to this regression: the computed gap is functional-dependent. Across the 7 molecules (gas and toluene) for which we ran both B3LYP/6-31G(d) and CAM-B3LYP/def2-TZVP, ΔE_{ST} differs by a mean 0.30 eV (up to 0.58 eV) and changes relative magnitude (Table S12); a higher-tier benchmark across six range-separated functionals on 27 emitters [31] shows the same effect at finer resolution: a median inter-functional spread of 0.05 eV on a quantity that approximates *half* the gap, hence roughly 0.1 eV on ΔE_{ST} itself. Any claim of sub-0.1 eV accuracy against such a reference is not physically meaningful. This second effect bounds what a *computed* gap can mean; it does not bound a model regressed on measured labels, and we do not add the two into a single floor. Only the label term applies directly to the present task. Among the representations tested here, larger learner capacity does not close the gap. Histogram gradient boosting reaches 0.105 eV, an MLP 0.169 eV and an RBF SVR 0.120 eV (Table S20); a second comparison over a differently configured set of learners places plain gradient boosting at 0.098 eV, within 0.002 eV of the forest and well below the proposed 0.017 eV comparison margin (Table S20 lists the histogram-boosting configuration). A frozen pretrained SMILES transformer (ChemBERTa, 77M parameters), the representation a current screening pipeline would reach for first, is no exception: its mean-pooled embeddings feed a random-forest or ridge head to MAE 0.109 eV to 0.113 eV on the same scaffold folds—it does not beat the 0.107 eV mean-value baseline, and is worse than Morgan by a paired 0.022 eV (95 % CI 0.008 to 0.034, excluding zero; `code/a013_pretrained_encoder_baseline.py`). Fine-tuning this encoder on only 231 labels was not tested here. Among the representations evaluated, added capacity did not improve the measured-gap error; this does not exclude other architectures or training sets. The negative lower R^2 bound is not a fingerprint-width artefact, since 512-, 1 024- and 2 048-bit Morgan models share it (Table S20).

Stratifying by label quality does not isolate a label effect. Restricting to molecules that carry replicates, the model reaches MAE 0.074 eV on the 79 whose reports agree to $\leq$ 0.05 eV, 0.106 eV on the 23 whose reports do not, and 0.107 eV on the 129 for which label precision cannot be measured. That ordering is not evidence about labels: a constant predictor fitted within each stratum reproduces it almost exactly (0.074, 0.107 and 0.110 eV), because the strata differ in the dispersion of the target itself (standard deviation 0.10, 0.15 and 0.20 eV). Within any one stratum the model carries no advantage over a constant. We therefore report no stratified evidence for a label ceiling; the quantitative statement that survives is the label-noise estimate above.

A related question is whether *uncontrolled measurement conditions*, rather than label imprecision as such, set the limit. The corpus records no solvent, host or temperature for individual values, so a strictly condition-controlled subset cannot be built; the closest available grouping is by source paper, without a guarantee that measurements in one publication share a laboratory or solvent. Splitting the benchmark on that (218 single-source against 13 multi-source molecules) and scoring the random forest on scaffold cross-validation folds, the model's descriptive advantage over a constant fitted within each full group is 0.0071 and 0.0024 eV, respectively. The latter comparator is not held out: a separate out-of-fold recalculation gives 0.1006 eV and 0.0693 eV for the single- and multi-source groups. The old group-difference interval (−0.014 to 0.035) used the in-sample constants and must not be interpreted as a matched predictive comparison. The raw errors run the other way (0.058 against 0.093 eV) but the multi-source group's target dispersion is half as large and its in-sample constant scores 0.061 eV; the held-out constant scores 0.069 eV instead. With only 13 molecules drawing on more than one source paper, neither a benefit nor a penalty of mixed source membership is resolved by these comparisons. The corpus does not permit a controlled test of solvent or measurement protocol.

A larger but differently processed corpus gives a stress test, not an isolated sample-size experiment. Featurising 497 additional experimental emitters through the same pipeline (a 728-molecule set) raises the MAE from 0.098 to 0.124 eV, drives R^2 from 0.18 to −0.01, and roughly halves the rank correlation (Spearman ρ from 0.34 to 0.16); the replicate-measured subset (n = 465) recovers only about half of the loss (Table S21). In this expanded, differently processed set, prediction worsens rather than improves. The added molecules were resolved through a different structure pipeline, so broader chemistry, target quality, and preprocessing are

confounded; the comparison cannot determine whether corpus size alone helps or hurts.

A separate concern is the descriptor side: the NTO features are computed at the *vertical* (ground-state) geometry, whereas the measured gap reflects relaxed excited states. We bounded this approximation by recomputing ΔE_{ST} adiabatically for 14 representative emitters in gas and toluene at CAM-B3LYP/def2-TZVP with excited-state optimisation (Table S14). Relative to these adiabatic references the vertical gap carries a mean absolute deviation of 0.195 eV, which exceeds the model's own error. The deviation is dominated by an additive offset rather than by reordering—a linear fit gives slope 1.00 and intercept 0.07 eV (R^2 = 0.70, Spearman ρ = 0.71)—so relaxation shifts the scale more than it scrambles the ranking the triage relies on. The residual is not negligible, however: the RMS deviation is 0.245 eV and the worst molecule departs by 0.548 eV, with sign changes between compounds, and 14 molecules cannot settle how much of that is systematic. We take the vertical descriptors as an acceptable ranking input on this evidence, not as a validated approximation to the adiabatic gap.

Candidate triage (enrichment 1.2–1.4× over random for low-gap emitters, established only at cutoffs of about 50 molecules or a fifth of the benchmark) and the 22 194-structure substructure-filtered library, where split-conformal intervals rule out single-molecule selection, are reported in SI section S6.7.

3.5. The cited TADF-gap models report error on computed gaps, not ranking across held-out publications

The low errors reported for TADF-gap models are obtained against computed targets: a TD-DFT gap inside a >1 000 000-molecule screen [7], semi-empirical gaps over a virtual library [32], ADC(2) gaps of inverted-gap molecules reproduced to 0.02 eV [8], quasiparticle-corrected Kohn–Sham gaps for 233 INVEST and TADF molecules at 0.06 eV [33], and ωB2PLYP′ gaps that Nigam et al. moved from regression to a good/bad classifier [9] (Table S23). He et al. [10] trained on 1 321 measured gap records with measurement-condition features but did not compare fingerprint and semi-empirical NTO representations under molecule, scaffold and source-publication holdout. Against measured gaps, whose labels carry about 0.072 eV RMS of inter-report noise, the Morgan and NTO models reach 0.091 eV to 0.096 eV, 10 % to 15 % below the mean-value predictor (naive baselines in Table S22). Whether the computed-gap estimators keep their accuracy against measured gaps has not been tested, and none of these studies reports how ranking holds when whole publications are held out, the test made in section 3.3.

3.6. Limitations

All results come from one corpus of 231 molecules, so the holdout-unit comparison does not test transfer to a new laboratory or a controlled protocol. The target is a consensus literature value (inter-report RMS scatter 0.072 eV) of a single scalar, ΔE_{ST}; solvatochromic, conformer-dependent and device-level quantities are not predicted. The NTO descriptors use ground-state vertical geometries (mean absolute deviation 0.195 eV from adiabatic references for 14 emitters), inverted-gap emitters are out of scope (both negative labels receive positive predictions), and only tabular learners and one frozen pretrained encoder were tested. Ranking is moderate ($\rho \approx 0.34$ to 0.36), so the models support triage, not selection. The full list is in SI section S6.8.

4. Conclusions

Do semi-empirical NTO descriptors retain rank information on measured TADF singlet–triplet gaps that structural fingerprints lose when molecules come from publications absent from training? In this corpus they do: under source-paper holdout the NTO model orders emitters across publications absent from training with median Spearman ρ = 0.34, against 0.17 for Morgan fingerprints. The contribution is a matched three-unit holdout comparison showing that fingerprint and RDKit ranking of measured TADF gaps depends on same-publication molecules in training, while semi-empirical NTO ranking does not. Neither representation orders molecules within a held-out paper, and neither has lower error than the other. Three results delimit that answer.

Fingerprint and NTO models do not differ in error.. Random forests built from the 2D structure alone—Morgan fingerprints (MAE = 0.091 eV) and RDKit descriptors (0.100 eV)—predict experimental ΔE_{ST} as well as NTO descriptors derived from a semi-empirical excited-state calculation (0.096 eV), against 0.107 eV for the mean-value predictor; the paired Morgan-minus-NTO MAE difference has a 95 % CI of −0.0171 to +0.0092 eV, and it stays near zero under paper holdout. Both the Morgan and NTO models sit only 10 % to 15 % below the mean. This null settles one design question: the sTDA calculation is not justified by accuracy, and its measurable value is ranking across publications. The 10 % to 15 % margin over the mean

is the bar a new structure representation has to clear on these labels.

The holdout unit, not the metric, sets the ranking contrast.. The NTO-minus-Morgan $\Delta\rho$ is 0.16 under source-paper holdout (2.5 % to 97.5 % span 0.07 to 0.25 over 50 fold assignments) against 0.073 under scaffold and 0.055 under molecule holdout. NTO ranks higher in 19 of 20 repeated paper partitions, but in only 11 and 13 of 20 matched scaffold and molecule partitions, so the 19/20 count belongs to the paper unit. The contrast comes from the fingerprint side: NTO ρ is nearly constant across units, while Morgan ρ falls from 0.305 to 0.172, and RDKit 2D descriptors fall with it (0.303 to 0.209), so among the three sets tested the invariance is specific to the NTO descriptors. Morgan ranks mainly molecules whose same-paper siblings are in training; on molecules from single-molecule papers its ρ is 0.10, and the NTO-minus-Morgan contrast on those molecules is not resolved (95 % CI −0.05 to 0.37). Siblings share both close structure and measurement protocol, so this is an association, not an identified cause. SHAP on the NTO variant ties the learned signal to hole–electron overlap and charge-transfer character; it is an interpretable association, not a causal validation.

Independent reports of the same compound scatter by 0.072 eV RMS. For the 102 multi-report molecules the median label has an RMS standard error of 0.049 eV; over all 231 labels the estimated noise is about 0.072 eV RMS, a measurable part of the error budget but not all of it. Across 231 molecules and 173 source papers, the report scatter supplies a direct estimate of measurement variability. Error is lower on molecules whose replicate reports agree (0.074 eV against 0.106 eV), but a constant predictor reproduces that ordering: it reflects different spreads of the target, not evidence that removing label noise alone improves the model. The estimated ceiling, $R^2_{\max} \approx 0.80$–0.86, lies well above the observed $R^2 \approx 0.26$. In the tested representation set a frozen pretrained molecular-transformer embedding reaches MAE 0.109 eV to 0.113 eV, worse than the Morgan model; other representations remain untested. Separately, the computed gap is functional-dependent (mean 0.30 eV, up to 0.58 eV between B3LYP and CAM-B3LYP); this concerns computed references, not the experimental-label noise estimate.

For modellers the consequence is practical. On literature-mined data, random and scaffold splits leave same-paper siblings in training and, in this corpus, overstate fingerprint ranking skill; a source-publication holdout should be reported with them before a representation is chosen for ranking. Where ranking on unseen publications matters, the semi-empirical NTO calculation is the representation that kept its rank information here: it orders emitters across publications absent from training, although no representation orders molecules within a held-out paper. As a triage filter the structure-based model enriches low-gap candidates 1.2 to 1.4× over random selection at broad cutoffs; at top-10 or top-20 the confidence intervals include random selection, and on the filtered library the conformal intervals are too wide for molecule-level selection, so no ranked shortlist is deposited.

All results come from one corpus of 231 molecules: the holdout-unit comparison does not test transfer to a new laboratory or a controlled protocol, and 128 of the 173 papers contribute a single molecule. The target is one scalar, ΔE_{ST}; the vertical NTO descriptors depart from adiabatic references by 0.195 eV on average for 14 emitters, and the data do not support molecule-specific spin–orbit-coupling, k_{RISC}, emission-wavelength or external-quantum-efficiency estimates (section 3.6).

What a controlled benchmark could resolve.. Label noise of about 0.072 eV RMS over the 231 molecules motivates single-source, solvent-controlled measurements, with documented host and protocol. Such a benchmark would separate measurement scatter from the remaining modelling headroom; it does not follow from these data that a better model could not improve on the current result.

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Declaration of competing interest

The authors declare no competing interests.

Author contributions

J.-P.T.N.: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. S.C.T.K.: Software, Validation, Writing – review & editing. P.S.M.K.: Validation, Writing – review & editing. P.S.: Resources, Validation, Writing – review & editing. S.G.N.E.: Supervision, Project administration, Writing – review & editing.

Data and code availability

The measured ΔE_{ST}labels come from the ChemDataExtractor-generated TADF database of Huang and Cole [15]. Data, code and the full analysis ledger for the three related articles are publicly archived under the Zenodo concept DOI 10.5281/zenodo.17436069, which resolves to the latest version, and are mirrored at `https://github.com/TchapetNjafa/Data_Articles_TADF`. The scripts and outputs of the source-paper holdout and holdout-unit analyses (ledger entries A-016 to A-020) are in that archive (`ML_reproducibility/code` and `ML_reproducibility/data`). They are also enclosed as Supplementary Data, in the folder `supplementary_data/` and the archive `JMGM_group_ranking_supplement.zip`. The files are, code: `jmgm_doi_group_ranking_check.py`, `_dataset.py`, `a017_split_unit_controls.py`, `a018_noise_ceiling_heterogeneous.py`, `a019_series_shortcut_test.py`, `a020_controls.py`, `fig_holdout_unit_ranking.py`; data: `a016_jmgm_doi_group_input.csv`, `a016_jmgm_doi_group_ranking.json`, `a017_split_unit_controls.json`, `a018_noise_ceiling_heterogeneous.json`, `a019_series_shortcut_test.json`, `a020_controls.json` and `a006_equivalence_ceiling.json`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

In preparing this work the authors used AI assistants (Claude by Anthropic — Sonnet 4.5, accessed May 2026; Sonnet 5 and Opus 5, accessed September 2026 — and ChatGPT by OpenAI) for literature triage, LaTeX formatting and consistency checking, drafting and revision of manuscript text, and analysis scripts used for paired comparisons, enrichment, label-precision audits and model baselines. The current revision (October 2026) used Claude Code (Opus 5.5 and Sonnet 5.5, Anthropic) for the holdout-unit control scripts, drafting and consistency checking, and OpenAI Codex (GPT-6) and GitLab Duo for independent review checks; every number was verified against the deposited analysis outputs. The graphical abstract was rendered by an AI image generator from an author-written specification. No AI system is listed as an author. The human authors retain responsibility for the final text, methods, figures and interpretation.

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References

[1] H. Uoyama, K. Goushi, K. Shizu, H. Nomura, C. Adachi, Highly efficient organic light-emitting diodes from delayed fluorescence, Nature 492 (7428) (2012) 234–238. doi:10.1038/nature11687.

[2] M. Yokoyama, K. Inada, Y. Tsuchiya, H. Nakanotani, C. Adachi, Trifluoromethane modification of thermally activated delayed fluorescence molecules for high-efficiency blue organic light-emitting diodes, Chemical Communications 54 (59) (2018) 8261–8264. doi:10.1039/c8cc03425g.

[3] B. Bugbee, Toward an optimal spectral quality for plant growth and development: The importance of radiation capture, Acta Horticulturae 1134 (2016) 1–12. doi:10.17660/ActaHortic.2016.1134.1.

[4] S. Curtarolo, G. L. W. Hart, M. B. Nardelli, N. Mingo, S. Sanvito, O. Levy, The high-throughput highway to computational materials design, Nature Materials 12 (2013) 191–201. doi:10.1038/nmat3568.

[5] K. T. Butler, D. W. Davies, H. Cartwright, O. Isayev, A. Walsh, Machine learning for molecular and materials science, Nature 559 (2018) 547–555. doi:10.1038/s41586-018-0337-2.

[6] E. O. Pyzer-Knapp, J. W. Pitera, P. W. J. Staar, S. Takeda, T. Laino, D. P. Sanders, J. Sexton, J. R. Smith, A. Curioni, Accelerating materials discovery using artificial intelligence, high performance computing and robotics, npj Computational Materials 8 (2022) 84. doi:10.1038/s41524-022-00765-z.

[7] R. Gómez-Bombarelli, J. Aguilera-Iparraguirre, T. D. Hirzel, D. Duvenaud, D. Maclaurin, M. A. Blood-Forsythe, H. S. Chae, M. Einzinger, D.-G. Ha, T. Wu, G. Markopoulos, S. Jeon, H. Kang, H. Miyazaki, M. Numata, S. Kim, W. Huang, S. I. Hong, M. Baldo, R. P. Adams, A. Aspuru-Guzik, Design of efficient molecular organic light-emitting diodes by a high-throughput virtual screening and experimental approach, Nature Materials 15 (2016) 1120–1127. doi:10.1038/nmat4717.

[8] L. Barneschi, L. Rotondi, D. Padula, Molecular geometry impact on deep learning predictions of inverted singlet–triplet gaps, J. Phys. Chem. A 128 (12) (2024) 2417–2426. doi:10.1021/acs.jpca.4c00172.

[9] A. Nigam, R. Pollice, P. Friederich, A. Aspuru-Guzik, Artificial design of organic emitters via a genetic algorithm enhanced by a deep neural network, Chem. Sci. 15 (7) (2024) 2618–2639. doi:10.1039/d3sc05306g.

[10] Z. He, H. Zhang, Y. Wang, W. Yang, J. Chen, H. Bi, Multiagent data mining for enhanced machine learning prediction of delayed fluorescence materials, ACS Omega 11 (31) (2026) 46279–46289. doi:10.1021/acsomega.6c03486.
URL http://dx.doi.org/10.1021/acsomega.6c03486

[11] F. Ferri, M. Cannariato, L. Pallante, E. A. Zizzi, M. Miceli, M. A. Deriu, Explainable machine learning and deep learning models for predicting TAS2R-bitter molecule interactions, J. Mol. Graph. Model. 142 (2026) 109187. doi:10.1016/j.jmgm.2025.109187.

[12] S. Noreen, M. J. Aljaafreh, A. M. Ahmed, Exploring the structural basis of organic compounds by predicting experimental IR peaks: A machine learning analysis, J. Mol. Graph. Model. 144 (2026) 109304. doi:10.1016/j.jmgm.2026.109304.

[13] J.-P. Tchapet Njafa, E. V. Kameni Tcheuffa, A. Maghame Foumkpou, S. G. Nana Engo, xtb-based high-throughput screening of tadf emitters: 747-molecule benchmark, Journal of Chemical Information and Modeling 66 (4) (2026) 2230–2248. doi:10.1021/acs.jcim.5c02978.

[14] J.-P. Tchapet Njafa, E. V. Kameni Tcheuffa, A. M. Foumkpou, S. G. Nana Engo, Data-driven design guidelines for tadf emitters from a high-throughput screening of 747 molecules, Journal of Chemical Information and Modeling (2026). doi:10.1021/acs.jcim.5c03068.

[15] D. Huang, J. M. Cole, A database of thermally activated delayed fluorescent molecules auto-generated from scientific literature with ChemDataExtractor, Scientific Data 11 (1) (Jan. 2024). doi:10.1038/s41597-023-02897-3.
URL http://dx.doi.org/10.1038/s41597-023-02897-3

[16] B. Das, A. Mondal, Physics-guided machine learning of excited-state properties for the design of high-performance tadf emitters, J. Mater. Chem. A 13 (2025) 33553–33570. doi:10.1039/D5TA03374H.

[17] A. Shuaib, L. Salah, A. Prlj, N. Rajendran, M. K. Etherington, A. A. Nazeer, C. S. Ponseca, A. P. Monkman, A. Danos, S. Makhseed, Singlet–triplet gaps for evaluating thermally activated delayed fluorescence: which one is the (b)E_{ST}?, J. Mater. Chem. C 14 (11) (2026) 4353–4361. doi:10.1039/d5tc03735b.

[18] D. Crusius, F. Cipcigan, P. C. Biggin, Are we fitting data or noise? analysing the predictive power of commonly used datasets in drug-, materials-, and molecular-discovery, Faraday Discuss. 256 (2025) 304–321. doi:10.1039/D4FD00091A.

[19] G. W. Bemis, M. A. Murcko, The properties of known drugs. 1. molecular frameworks, Journal of Medicinal Chemistry 39 (15) (1996) 2887–2893. doi:10.1021/jm9602928.

[20] Y. Olivier, J.-C. Sancho-García, L. Muccioli, G. D'Avino, D. Beljonne, Computational design of thermally activated delayed fluorescence materials: The challenges ahead, The Journal of Physical Chemistry Letters 9 (20) (2018) 6149–6163. doi:10.1021/acs.jpclett.8b02327.

[21] C. Bannwarth, S. Ehlert, S. Grimme, GFN2-xTB—an accurate and broadly parametrized self-consistent tight-binding quantum chemical method with multipole electrostatics and density-dependent dispersion contributions, Journal of Chemical Theory and Computation 15 (3) (2019) 1652–1671. doi:10.1021/acs.jctc.8b01176.

[22] S. Grimme, C. Bannwarth, Ultra-fast computation of electronic spectra for large systems by tight-binding based simplified Tamm-Dancoff approximation (sTDA-xTB), The Journal of Chemical Physics 145 (7) (2017) 074108. doi:10.1063/1.4991798.

[23] R. L. Martin, Natural transition orbitals, The Journal of Chemical Physics 118 (11) (2003) 4775–4777. doi:10.1063/1.1558471.

[24] G. Landrum, et al., RDKit: Open-source cheminformatics, https://www.rdkit.org, version 2026.03 (2016).

[25] L. Breiman, Random forests, Machine Learning 45 (1) (2001) 5–32. doi:10.1023/A:1010933404324.

[26] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, E. Duchesnay, Scikit-learn: Machine learning in Python, Journal of Machine Learning Research 12 (2011) 2825–2830.

[27] F. Neese, F. Wennmohs, U. Becker, C. Riplinger, The ORCA quantum chemistry program package, The Journal of Chemical Physics 152 (22) (2020) 224108. doi:10.1063/5.0004608.

[28] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, Advances in Neural Information Processing Systems 30 (2017) 4765–4774.

[29] E. Salvi, G. Agostini, S. Veglianti, G. J. Costa, L. De Vico, D. Padula, C. A. Guido, On the role of electronic correlation and state-specific environment polarization in singlet–triplet gap inversion, J. Comput. Chem. 46 (30) (2025) e70267. doi:10.1002/jcc.70267.

[30] L. Dang, et al., A practical guide to measuring and reporting photophysical data, Dalton Trans. (2025).

[31] T. Froitzheim, L. Kunze, S. Grimme, J. M. Herbert, J.-M. Mewes, Benchmarking charge-transfer excited states in TADF emitters: ΔDFT outperforms TD-DFT for emission energies, J. Phys. Chem. A 128 (30) (2024) 6324–6335. doi:10.1021/acs.jpca.4c03273.

[32] K. Thapa, J. I. Jones, L. E. Ratcliff, A high throughput virtual screening approach for identifying new thermally activated delayed fluorescence-based emitters, J. Mater. Chem. C 13 (37) (2025) 19212–19225. doi:10.1039/d5tc01938a.

[33] Y. Li, C. Li, Efficient prediction of singlet–triplet gaps in delayed fluorescence emitters using Kohn–Sham orbital energies, J. Phys. Chem. Lett. 17 (1) (2026) 141–148. doi:10.1021/acs.jpclett.5c03449.